21. The time until first failure of a brand of inkjet printers is normally distributed with a mean of 1,500 hours and a standard deviation of 200 hours. What proportion of printers fails before 1,000 hours?

(a) 0.0062 (b) 0.0668

(c) 0.8413 (d) 0.0228

22. The weights of newborn human babies are normally distributed with a mean of 3.2kg and a standard deviation of 1.1kg. What is the probability that a randomly selected newborn baby weighs less than 2.0kg?

(a) 0.138 (b) 0.428

(c) 0.766 (d) 0.262

23. Monthly expenditure on their credit cards, by credit card holders from a certain bank, follows a normal distribution with a mean of ₹1,295.00 and a standard deviation of ₹750.00. What proportion of credit card holders spend more than ₹1,500.00 on their credit cards per month?

(a) 0.487 (b) 0.392

(c) 0.500 (d) 0.791

24. Let z be a standard normal variable. If the area to the right of z is 0.8413, then the value of z must be:

(a) 1.00 (b) –1.00

(c) 0.00 (d) –0.41

25. If the area to the left of a value of z (z has a standard normal distribution) is 0.0793, what is the value of z?

(a) –1.41 (b) 1.41

(c) –2.25 (d) 2.25

26. If $P(Z > z) = 0.8508$ what is the value of z (z has a standard normal distribution)?

(a) –0.48 (b) 0.48

(c) –1.04 (d) 1.04

27. If $P(Z > z) = 0.5832$ what is the value of z (z has a standard normal distribution)?

(a) –0.48 (b) 0.48

(c) 1.04 (d) –0.21

28. In a binomial distribution, the probability of success is twice as that of failure. Then out of 4 trials, the probability of no success is

(a) 16/81 (b) 1/16

(c) 2/27 (d) 1/81

Miscellaneous Problems

1. A manufacturer of metal pistons finds that on the average, 12% of his pistons are rejected because they are either oversize or undersize. What is the probability that a batch of 10 pistons will contain

   (a) no more than 2 rejects?

   (b) at least 2 rejects?

2. Hospital records show that of patients suffering from a certain disease 75% die of it. What is the probability that of 6 randomly selected patients, 4 will recover?

3. If electricity power failures occur according to a Poisson distribution with an average of 3 failures every twenty weeks, calculate the probability that there will not be more than one failure during a particular week.

4. Vehicles pass through a junction on a busy road at an average rate of 300 per hour.

   1. Find the probability that none passes in a given minute.

   2. What is the expected number passing in two minutes?

5. Entry to a certain University is determined by a national test. The scores on this test are normally distributed with a mean of 500 and a standard deviation of 100. Raghul

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